

RAPID DEVELOPMENT AND IMPLEMENTATION OF A CUSTOM PHOTON STOP AT THE ADVANCED LIGHT SOURCE: FROM FAILURE ANALYSIS TO PREDICTIVE MAINTENANCE

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Abstract

In June of 2023, the Advanced Light Source (ALS) at Lawrence Berkeley National Laboratory (LBNL) in Berkeley, California, USA, experienced a vacuum interlock event that triggered a beam dump. Upon investigation, technicians discovered a leak in the cooling system of a custom photon stop (PS) in Sector 5 of 12. This paper will detail the event, the temporary restoration of operations, and how a new PS was designed, analyzed, fabricated, assembled, tested, qualified, installed, and commissioned in a 14-week window. The original PS had been in service for over 20 years when failure occurred. During this period, the manufacturing landscape changed significantly, with many specialized capabilities—particularly brazing techniques—becoming unavailable both locally and regionally. Schedule and fabrications constraints strongly drove the design of the replacement PS. Failure analysis of the original PS indicated that corrosion and blocking of coolant channels had led to thermal cycling-induced fatigue cracking in the top surface. A tool for predicting fatigue life of the new PS was developed to estimate remaining cycles to failure, providing operations staff a basis for predictive maintenance and advance warning to prevent unexpected failures in the future.

INTRODUCTION

The photon stop protects three beamlines at the ALS: 5.0.1, 5.0.2, and 5.0.3. These beamlines are used for macromolecular crystallography by the Berkeley Center for Structural Biology. They are fed by a wiggler with a minimum energy of 5,000 eV and a maximum energy of 16,000 eV in operation [1].

Failure of Original Photon Stop

The original PS had been installed and had operated successfully since ~1996. In June of 2023, a failure occurred in the top surface of the PS, close to the area of highest heat flux from the beam. A small crack had formed in the 0.065" thick faceplate, leading to the leaking of water coolant into the vacuum chamber, which tripped a vacuum interlock and triggered a beam dump, see Fig. 1.

Temporary Restoration of Operations

Upon detection of failure, the PS was immediately removed and temporarily repaired by TIG welding of the Glidcop AL-15 faceplate. This repair allowed the ALS storage ring to remain in operation but required that the gap in the BL5.0 wiggler remain fully open, limiting beam power and preventing the use of all three 5.0 beamlines. A

complete repair was planned to coincide with an upcoming ALS summer shutdown.

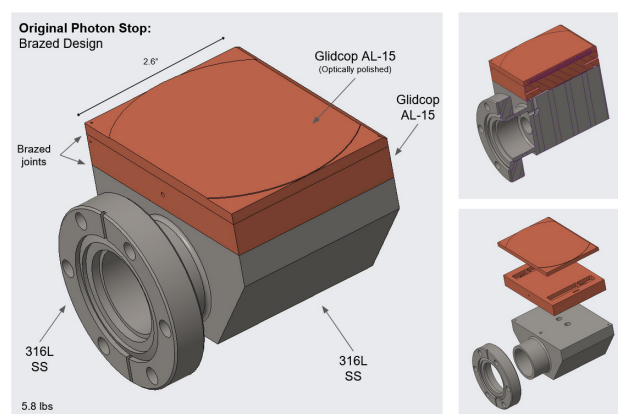


Figure 1: Design of original photon stop. Water coolant follows an anastomosing path through parallel courses of 0.5 mm wide channels below the top face.

DEVELOPMENT OF NEW PHOTON STOP

A replacement PS was developed to replace the temporarily repaired unit such that the beamline could be run at full power. The requirements were to block all radiation from the beamline and absorb and dissipate related power (~11.3 kW) while maintaining safe temperatures and stresses during operation. The original PS served as an IR mirror for a since-decommissioned beamline, and this was no longer a requirement, obviating related requirements for dimensional stability and optical surface quality. The minimum intended lifetime of the PS was ~3 years, leading up to an intended replacement of the entire ALS storage ring.

Schedule and Fabrication Constraints

Aggressive schedule constraints generated additional fabrication constraints, and these strongly drove the design of the new PS. While the original PS had a brazed design, it was determined early on that the timeline did not allow for the long lead-times brazing would require, particularly as on-site and regional machine shops no longer supported the kinds of complex brazing that were required. A wide range of alternative designs were investigated. For example, designs with incidence angles of 15° were investigated. These had the advantage of decreased maximum heat fluxes (15.9 W/mm² compared to 30.6 W/mm²), but fitment concerns and concerns related to excessive pressure drop in the significantly longer coolant channels precluded their further development. Ultimately, a design similar to that of the original PS was chosen, the

primary difference being that the new design employed TIG welding in place of brazing. Unlike brazing, TIG welding could be completed with existing resources and skilled personnel at the LBNL machine shops.

Design

The space envelope, angle of incidence (30°), hot wall thickness (0.065"), and approximate weight (~6.0 lbs) of the PS was maintained to avoid fitment complications and limit unintended consequences (Fig. 2).

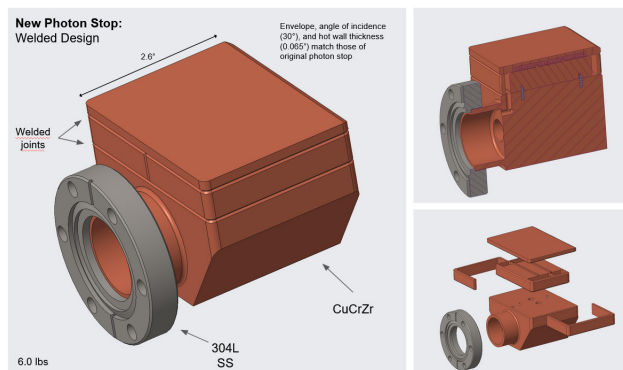


Figure 2: Design of replacement photon stop.

CuCrZr was chosen in place of Glidcop AL-15, given improved weldability, higher tensile strength, and similar thermal properties. The tensile strength, yield strength, and hardness of CuCrZr depend significantly on prior heat-treatment, as the material is a precipitation hardened alloy. See AL-1687-0575 [2] for details on the effects of welding and heat-treatment on CuCrZr.

Analysis

Steady-state thermal and static structural FEA modelling of the new PS and the original PS were completed. In addition, steady-state thermal analysis of an associated infrared mirror was completed to ensure radiation reflected by the new PS was safely dissipated.

The details and results of the analyses are given in AL-1686-7772 [3]. Reflected power was calculated to be ~5 W, and analysis of the infrared mirror indicated that associated temperatures and stresses were well within acceptable bounds.

With the new PS, while higher temperatures were observed, lower stresses and higher safety factors were obtained due to higher material strength and freedom of the material to deform (given welded joints across only sections of faces, as opposed to brazed joints across the entirety of faces). The minimum yield stress factor of safety of the new PS (0.69) was larger than that of the original PS (0.48). As with the original PS, it is expected that water in the coolant channels will temporarily boil in a hot central region and that material in the faceplate will plastically deform during temperature cycling. The implications of plastic deformation for fatigue-related failure are described in a later section.

Fabrication

Fabrication of the production units required ~4.5 weeks for CNC machining and ~3 weeks for welding. Electron beam welding was also considered, as this had successfully been employed for CuCrZr/SS welds on recent ALS projects, but long lead times made this approach unworkable. TIG welding proved challenging but successful. Delays were experienced due to the large number of sequential steps: Each of the four major rounds of welding required significant cleaning before welding and UHV leak testing after welding. Cleaning, welding, and leak checking each relied on singular skilled personnel, and delays were experienced due to availability of staff. Additional challenges were created by high levels of oxidation in the CuCrZr due to welding at elevated temperatures in open air, with increased risks of impurities and leaks.

Testing, Qualification, Installation, and Commissioning

The PS was tested and qualified, with hydrostatic pressure testing (3 successful tests with alcohol at 150 psi for 5 minutes per test) and coolant flow rate measurement (> 2 GPM). Installation and commissioning included Equipment Protection System testing, vacuum conditioning, and monitoring of coolant temperatures and vacuum pressures as beam power was increased to full power. The coolant temperature rise showed a steady increase with increasing total beam power, and no evidence of critical heat flux being reached was detected.

FAILURE ANALYSIS

Destructive analysis of the original PS supported the hypothesis that failure was due to thermally induced fatigue. By inspection, significant corrosion at areas of high temperature and turbulent flow caused dislodged debris to block some of the cooling channels. This was exacerbated by the coolant channels and intervening ribs of material having matching widths, which promoted jamming of debris in the channels (Fig. 3). Also, deposition of a layer of unknown material in the channels both blocked flow and insulated the part from the coolant, particularly in areas of highest beam intensity (Fig. 3).

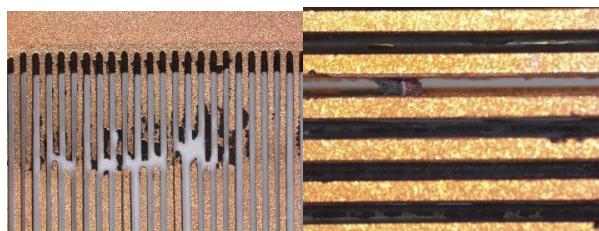


Figure 3: Significant corrosion occurred at areas of high temperature and turbulent flow, right. Dislodged debris and a layer of deposited material blocked flow and insulated the part from coolant, left.

Lack of coolant flow in blocked channels appears to have created conditions of thermally induced fatigue, resulting in cracking of the face near the area of highest

beam power, with one particularly large crack being visible at the site of failure (Fig. 4).



Figure 4: A vertical slice through the hot-wall of the original photon stop, showing cracking at the site of failure. Artifacts of the welded repair can be seen at top.

PREDICTING FATIGUE FAILURE

Given the risk of repeat failure by fatigue in the new PS, fatigue analysis was conducted using the results of the steady-state thermal and static structural finite element analysis described above, and this was parametrized by coolant flow rate, with results ranging from 560 cycles to failure at 0.5 GPM to 11,747 cycles to failure at 3.0 GPM.

Neuber analysis was used to calculate the cyclic response of the material to the minimum (compressive) principal stress determined in the finite element analysis, followed by unloading. Hysteresis loops were calculated to determine the resulting strain amplitude. The empirical strain-life expression for CuCrZr from [4] was then used to predict the number of cycles until cracks will begin to appear. Uniaxial fatigue, constant material properties, and zero creep were assumed. The results are shown in Fig. 5. A coolant flow rate of > 2.0 GPM was measured in-situ for the new PS, corresponding to an expected number of cycles to failure of $> 6,740$.

These results provide a useful tool for monitoring the fatigue lifetime of the PS. The expected number of cycles to failure is a function of the coolant flow rate, and monitoring the coolant flow rate to ensure sufficient expected cycles to failure is therefore recommended. Furthermore, given uncertainties in the fatigue analysis, a generous safety factor (e.g. $FOS = 2$) is recommended to determine a safe number of operating cycles. It is also recommended that both the faceplate surface and the weld joints be inspected for fatigue-related issues well before reaching the predicted number of cycles to failure.

BL5.0 New Photon Stop Estimated Cycles to Failure

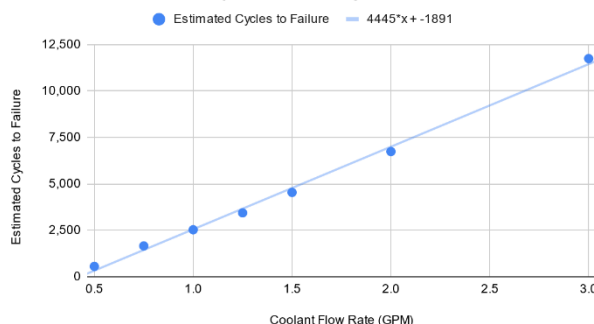


Figure 5: Estimated cycles to failure by fatigue in the top surface of the new photon stop faceplate.

CONCLUSIONS

The project's success demonstrated effective engineering response to critical infrastructure failure, delivering a high-performance replacement component within tight schedule constraints while maintaining the reliability standards required for synchrotron radiation facilities.

One cause of the original failure was that no analytical tools were available or protocols in place for assessing the lifetime of the original PS. Moving forward, a tool for predicting fatigue life of the new PS was developed to provide a basis for preventative maintenance.

For other projects facing similar requirements and constraints, the following are recommended:

- If schedule constraints allow, investigate the potential for electron beam welding CuCrZr and stainless steel to take advantage of possible benefits: smaller and less intense heat-affected zones, lower risks of weld failures, and higher repeatability.
- Investigate additive manufacturing (e.g. direct metal laser sintering of Cu alloys) to take advantage of possible benefits, such as potentially rapid production of parts with difficult-to-machine features such as conformal cooling channels. With appropriate heat-treatment, these materials may also be suitable for use in conflat knife edge seals, opening up possibilities for complex, monolithic, single-material parts [2].
- If TIG welding is employed, 1) design assemblies such that the number of sequential welding, cleaning, and testing steps is minimized and, 2) fabricate extra parts/assemblies to mitigate the risk of weld failures.
- Design channels with fin widths smaller than intervening channel widths to reduce the risk of dislodged pieces jamming in the channels.
- Investigate ways to adjust water chemistry to minimize corrosion (e.g. by minimizing concentrations of oxygen and/or sulphur).
- Install water filters both upstream and downstream of components experiencing high heat loads, allowing for both blockage protection and corrosion monitoring.

REFERENCES

- [1] N. Wenner *et al.* “BL5.0 Photon Stop Overview”, Lawrence Berkeley National Laboratory, Berkeley, CA, USA, Rep. AL-1686-7773, Dec. 2023.
- [2] N. Wenner *et al.* “Effects of Welding and Heat Treatment on Traditional and Additively Manufactured CuCrZr for UHV Applications”, Lawrence Berkeley National Laboratory, Berkeley, CA, USA, Rep. AL-1687-0575, Nov. 2023.
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- [4] P. Marmy, "Creep-fatigue of CuCrZr: A review of the existing fatigue, creep and creep-fatigue data base and a life prediction analysis using a time based damage evaluation", École Polytechnique Fédérale de Lausanne, Jan. 2005.